```
import math

board = [' ' for _ in range(9)]

def print_board():
  for i in range(3):
    print(board[3*i], '|', board[3*i+1], '|', board[3*i+2])
    if i < 3:
     print('---------')

def is_winner(b, player):
  wins = [
      (0,1,2),(3,4,5),(6,7,8),
      (0,3,6),(1,4,7),(2,5,8),
      (0,4,8),(2,4,6)
  ]
  for a, b_, c in wins:
      if b[a] == b[b_] == b[c] == player:
        return True
  return False

def is_board_full(b):
  return ' ' not in b

def minimax_alpha_beta(b, depth, alpha, beta, is_maximizing):

  if is_winner(b, 'X'):
    return -10 + depth
  if is_winner(b, 'O'):
    return 10 - depth
  if is_board_full(b):
    return 0

  if is_maximizing:
    max_eval = -math.inf
    for i in range(9):
      if b[i] == ' ':
         b[i] = 'X'
         eval = minimax_alpha_beta(b, depth + 1, alpha, beta, False)
         b[i] = ' '
         max_eval = max(max_eval, eval)
         alpha = max(alpha, eval)
         if beta <= alpha:
           break
    return max_eval
  else:
    min_eval = math.inf
    for i in range(9):
      if b[i] == ' ':
        b[i] = 'O'
        eval = minimax_alpha_beta(b, depth + 1, alpha, beta, True)
        b[i] = ' '
        min_eval = min(min_eval, eval)
```

```
        min_eval = min(min_eval, eval)
        beta = min(beta, eval)
        if beta <= alpha:
          break
    return min_eval

def best_move():
  best_score = -math.inf
  move = -1
  for i in range(9):
    if board[i] == ' ':
      board[i] = 'X'
      score = minimax_alpha_beta(board, 0, -math.inf, math.inf, Fals
      board[i] = ' '
      if score > best_score:
        best_score = score
        move = i
  return move

def play_game():
  print("Board positions are numbered 0 to 8:")
  print("0 | 1 | 2")
  print("---------")
  print("3 | 4 | 5")
  print("---------")
  print("6 | 7 | 8")
  print()

  while True:
    move = best_move()
    if move == -1:
      print("Game over. It's a tie!")
      break
    board[move] = 'X'
    print("AI (X) plays move:", move)
    print_board()
    if is_winner(board, 'X'):
      print("AI (X) wins!")
      break
    if is_board_full(board):
      print("Game Over! It's a tie!")
      break

    while True:
      try:
        human_move = int(input("Enter your move (0-8): "))
        if 0 <= human_move <= 8 and board[human_move] == ' ':
          break
        else:
          print("Invalid move. Try again.")
      except ValueError:
        print("Invalid input. Try again.")

    board[human_move] = 'O'
    print_board()
```

```
    print_board()
    if is_winner(board, 'O'):
      print("YOU WIN!!!")
      break
    if is_board_full(board):
      print("Game Over! It's a tie!")
      break

if __name__ == '__main__':
  play_game()
```

```
Board positions are numbered 0 to 8:
0 | 1 | 2
---------
3 | 4 | 5
---------
6 | 7 | 8

AI (X) plays move: 4
  |   |
---------
  | X |
---------
  |   |
---------
Enter your move (0-8): 0
O |   |
---------
  | X |
---------
  |   |
---------
AI (X) plays move: 8
O |   |
---------
  | X |
---------
  |   | X
---------
Enter your move (0-8): 2
O |   | O
---------
  | X |
---------
  |   | X
---------
AI (X) plays move: 5
O |   | O
---------
  | X | X
---------
  |   | X
---------
Enter your move (0-8): 1
O | O | O
---------
  | X | X
---------
  |   | X
```

```
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YOU WIN!!!
```